RAJALAKSHMI ENGINEERING COLLEGE

Prasanna K 251001187-2025-IT

P2

Started on	Friday, 11 September 2026, 12:29 PM
State	Finished
Completed on	Friday, 11 September 2026, 12:44 PM
Time taken	15 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

// Function to swap two elements
void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}

// Partition function
int partition(int arr[], int low, int high)
{
    int pivot = arr[high];
    int i = low - 1;

    for (int j = low; j < high; j++)
    {
        if (arr[j] <= pivot)
        {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }

    swap(&arr[i + 1], &arr[high]);

    return i + 1;
}

// Quick Sort function
void quickSort(int arr[], int low, int high)
{
    if (low < high)
    {
        // Find pivot position
        int pi = partition(arr, low, high);

        // Sort left part
        quickSort(arr, low, pi - 1);

        // Sort right part
        quickSort(arr, pi + 1, high);
    }
}

```

```
47 // Function to print the array
48 void printArray(int arr[], int n)
49 {
50     for (int i = 0; i < n; i++)
51         printf("%d ", arr[i]);
52
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Friday, 21 August 2026, 12:53 PM
State	Finished
Completed on	Friday, 11 September 2026, 12:44 PM
Time taken	20 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c

#include <stdio.h>

// Recursive function to find two elements whose sum is x
int findPair(int arr[], int left, int right, int x)
{
    // Base condition
    if (left >= right)
        return 0;

    int sum = arr[left] + arr[right];

    // Pair found
    if (sum == x)
    {
        printf("%d\n%d",arr[left], arr[right]);
        return 1;
    }

    // If sum is greater than x,
    // reduce the larger element
    if (sum > x)
        return findPair(arr, left, right - 1, x);

    // If sum is less than x,
    // increase the smaller element
    return findPair(arr, left + 1, right, x);

}

int main()
{
    int n;
    scanf("%d",&n);
    int arr[n] ;
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int x;
scanf("%d",&x);
    if (!findPair(arr, 0, n - 1, x))
        printf("No\n");

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Friday, 11 September 2026, 12:26 PM
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Completed on	Friday, 11 September 2026, 12:39 PM
Time taken	13 mins 19 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x)
{
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (arr[mid] == x)
        return arr[mid];

    if (arr[mid] > x)
        return findFloor(arr, low, mid - 1, x);

    int floorValue = findFloor(arr, mid + 1, high, x);

    if (floorValue != -1)
        return floorValue;

    return arr[mid];
}

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    int x;
    scanf("%d", &x);

    int result = findFloor(arr, 0, n - 1, x);

    if (result == -1)
        printf("Floor does not exist");
    else
        printf("%d", result);

    return 0;
}

```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Friday, 11 September 2026, 12:23 PM
State	Finished
Completed on	Friday, 11 September 2026, 12:39 PM
Time taken	16 mins 12 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int countOccurrences(int nums[], int low, int high, int target)
{
    int count = 0;

    for (int i = low; i <= high; i++)
    {
        if (nums[i] == target)
        {
            count++;
        }
    }

    return count;
}

int majorityElement(int nums[], int low, int high)
{
    if (low == high)
    {
        return nums[low];
    }

    int mid = low + (high - low) / 2;

    int leftMajority = majorityElement(nums, low, mid);
    int rightMajority = majorityElement(nums, mid + 1, high);

    if (leftMajority == rightMajority)
    {
        return leftMajority;
```

```
    }

    int leftCount = countOccurrences(nums, low, high, leftMajority);
    int rightCount = countOccurrences(nums, low, high, rightMajority);

    if (leftCount > rightCount)
    {
        return leftMajority;
    }
    else
    {
        return rightMajority;
    }
}

int main()
{
    int n;
    scanf("%d", &n);

```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Started on	Friday, 11 September 2026, 12:22 PM
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Completed on	Friday, 11 September 2026, 12:39 PM
Time taken	16 mins 50 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n, a[100];
    int low = 0, high, mid, first;

    scanf("%d", &n);

    for(int i = 0; i < n; i++){
      scanf("%d", &a[i]);
}
    high = n - 1;
    first = n;

    while(low <= high)
    {
        mid = (low + high) / 2;

        if(a[mid] == 0)
        {
            first = mid;
            high = mid - 1;
        }
        else
        {
            low = mid + 1;
        }
    }

    printf("%d", n - first);

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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